

Project Title

Zomato Restaurant Data Analysis

Objective

The objective of this project was to analyze the Zomato restaurant dataset to identify trends in restaurant ratings, cuisines, locations, pricing, and customer preferences using Python and data visualization techniques.

Dataset

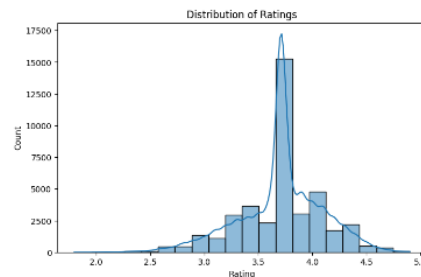
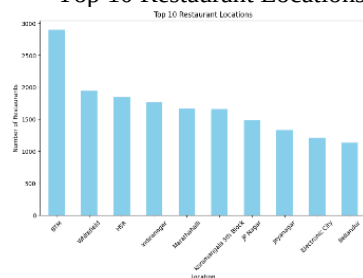
- Dataset: Zomato Restaurant Dataset
- Total Records: 56,252
- Total Features: 13

Data Cleaning

- Removed duplicate records.
- Handled missing values.
- Converted rating, votes, and cost columns into numeric format.
- Cleaned text fields for analysis.

Visualizations Created

- Top 10 Restaurant Locations



- Top 10 Cuisines
- Ratings Distribution
- Price vs Rating Scatter Plot
- Correlation Heatmap
- Cuisine Word Cloud

Key Findings

- Most restaurants are concentrated in a few major locations.
- North Indian cuisine is one of the most popular cuisines.
- Most restaurants have ratings between 3.5 and 4.2.
- Restaurant price does not always determine customer ratings.
- Higher customer votes generally indicate more popular restaurants.

Recommendations

1. Partner with highly rated restaurants.
2. Promote popular cuisines to attract more customers.
3. Recommend restaurants based on user location.
4. Highlight affordable restaurants with high ratings.
5. Use ratings and customer votes to improve recommendations.

Tools Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- WordCloud

Conclusion

This project successfully analyzed restaurant data using data cleaning, exploratory data analysis, and visualization techniques. The insights obtained can help improve restaurant recommendations and business decision-making.